THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS CHAPTER TEST SET A

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A linear equation in two variables can be written in the standard form ax + by + c = 0, where: 1

(a) a and b are not both zero **(b)** a and b are both zero

(c) c must be non-zero **(d)** a and b must be integers

2. The slope of the line y = −4 is: 1

(a) 4 **(b)** −4 **(c)** undefined **(d)** 0

3. **Assertion (A):** The point (0, 0) is a solution of 3x − 5y = 0. 1

Reason (R): Whenever c = 0 in ax + by + c = 0, the values x = 0, y = 0 satisfy the equation.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

4. **Assertion (A):** The line y = 3x + 2 passes through the point (0, 2). 1

Reason (R): The slope of the line y = 3x + 2 is 3.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

5. If x and y satisfy 2x + y = 12 and x + 3y = 11, then the value of x + y is: 1

(a) 8 **(b)** 10 **(c)** 7 **(d)** 5

6. The slope of the line through the points (-2, 5) and (4, -1) is: 1

(a) 1 **(b)** -2 **(c)** -1 **(d)** 0

7. The slope of the line 3x − 6y = 12 is: 1

(a) 3/2 **(b)** -2 **(c)** 1/2 **(d)** -1/2

8. Which of the following points lies on the line 5x − 2y = 4? 1

(a) (4, 0) **(b)** (1, 3) **(c)** (2, 3) **(d)** (2, 2)

9. The pair of equations x + 2y = 5 and 2x − y = 3 has: 1

(a) a unique solution **(b)** no solution

(c) exactly two solutions **(d)** infinitely many solutions

10. The point (1, k) lies on the line 3x + 2y = 7. The value of k is: 1

(a) 2 **(b)** 1 **(c)** 4 **(d)** 3

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Write 3y = 5x − 7 in the form ax + by + c = 0 and state the values of a, b and c. 2

12. Is $(3, -2)$ a solution of $4x + 5y = 2$? Justify your answer. 2

13. Find two solutions of $7x - 2y = 14$. 2

14. If $(2, -1)$ is a solution of $3ax + 2y = 4$, find a. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by the substitution method: $2x + y = 11$ and $x - 3y = -5$. 3

16. Solve by the elimination method: $3x + 4y = 18$ and $5x - 2y = 4$. 3

17. 3 kg apples and 2 kg grapes cost ₹540, while 2 kg apples and 5 kg grapes cost ₹690. Find the price per kg of each fruit. 3

OR

A cab company charges a fixed fare plus a fixed rate per km. For 8 km the fare is ₹140 and for 12 km it is ₹200. Find the fixed fare, the rate per km and the fare for 20 km. 3

18. The sum of the digits of a two-digit number is 11. The number is 27 more than the number formed by reversing its digits. Find the number. 3

OR

A father is 4 times as old as his son. After 5 years, he will be 3 times as old as his son. Find their present ages. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Solve the pair $3x - 5y = -1$ and $2x + 3y = 12$ by the elimination method, and verify the solution in both equations. 5

OR

Solve graphically: $x + 2y = 8$ and $2x - y = 1$. Prepare a table for each line, plot the lines, read the solution and state its nature. 5

20. For the pair $kx + 4y = k - 4$ and $16x + ky = k$, find the value of k for which the pair has no solution, and describe the graphs. 5

OR

Two lines $y = \frac{1}{2}x + 2$ and $y = -x + 5$ are drawn. 5

(i) Write the slope and y-intercept of each.

(ii) Find their point of intersection algebraically.

(iii) Say what the slopes tell about the number of solutions.

END OF QUESTION PAPER — Total: 40 Marks